

PEDRAM MOHAMMADI

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PROFESSIONAL SUMMARY

- Expert knowledge and understanding in the areas of:
 - Subjective and objective image/video quality assessment, High Dynamic Range (HDR) video processing, Computational photography, Statistical analysis, Image segmentation, Tone mapping and Inverse tone mapping algorithms, Perceptual transfer functions, Human Visual System (HVS), Color processing and gamuts, Convex optimization, Video encoding, Video rate control
- Advanced programming skills using Python and MATLAB. Basic understanding of C programming.
- Familiar with FFMPEG, Linux, Git, Davinci Resolve, Elecard StreamEye

EDUCATION

Ph.D. in Electrical and Computer Engineering

2015 - 2020

University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering

2012 - 2014

Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer

2021 - Present

NETINT Technologies Inc., Canada

- Designed and developed a patented tools for enhancing the visual quality of NETINT's next-generation VPUs, resulting in 15% increase in objective video quality while saving bit-rate.
- Designed and developed patented video rate control algorithms for NETINT's next generation VPUs resulting in 10% BD-rate improvement.
- Implemented various video post-processing algorithms to enhance VMAF objective quality score of NETINT's next generation VPUs.
- Designed and developed a patented real-time SDR/HDR conversion suite for NETINT's next-generation VPUs.
- Conducted extensive fine-tuning experiments for various company products resulting in an average of 5% increase in objective visual quality.
- Mentored junior engineers and co-op students, providing ongoing support to meet project goals.

Scientific Research and Experimental Development (SR&ED) Consultant 2020 - 2021
KPMG LLP, Canada

- Successfully increased SR&ED tax credits for clients by an average of 20-30% annually, resulting in a total of 500K to 2 million dollars in additional credits and refunds over 1.5 years.
- Improved client cash flow by facilitating the timely receipt of SR&ED refunds, contributing to a 10-15% increase in reinvestment funds for R&D activities.

Research Scientist 2015 - 2020
The University of British Columbia, Canada

- Designed and patented an SDR-to-HDR conversion algorithm, enhancing subjective visual quality by an average of 80%.
- Devised an image segmentation algorithm based on entropy theory capable of dividing a frame to various brightness regions.
- Developed a color adjustment algorithm, improving color accuracy by an average of 70%.
- Designed a flickering reduction algorithm for visual stability.
- Authored multiple research papers in prestigious IEEE journals, and conferences.

Application Developer I 2018 - 2019
TELUS Communications Inc., Canada

- Led a team to implement an SDR to HDR conversion system, achieving real-time performance for 1080p content.
- Developed high-quality code, enhancing product performance and user experience.

PATENTS

- **P. Mohammadi**, J. Li, E. Andrade Neto, and J. Lou "Methods and Apparatus for video rate control," Patent filed, filing date: January 2024.
- **P. Mohammadi**, E. Andrade Neto, and J. Lou "Image processing methods, devices, electronic devices and storage media," China Patent Office, Application No. CN116167950B, Publication date: April 2023.
- P. Nasiopoulos, S. Ploumis, M. T. Pourazad, R. Boitar, and **P. Mohammadi**, "Methods and Apparatuses for Tone Mapping and Inverse Tone Mapping," U.S. Patent, Application No. US11100888B2, Publication date: January 2019.

PUBLICATIONS

For a complete list of my publications please visit: [My Google scholar page](#)

WORK AUTHORIZATION

Currently holding a Canadian citizenship